

# Benchmarking of NQCH's quantum computer

March 19, 2026

## 1. Report of Changes

**Platform:** sinq20

**Calibration-id:** 44a690428e4e7e478965661e456986232914ef40

**Calibration date:** 2026-02-26 16:38:55

**Calibration note:** chore(sinq20): update readout line B readout classifiers

**Experiment-id:** 20260319212202

**Experiment date:** 2026-03-19 21:22:03

**Experiment note:** temporary note!!!

**Platform:** sinq20

**Calibration-id:** 4dc4082f38a53222b3956c22202d32a520d4bc78

**Calibration date:** 2026-01-15 02:03:03

**Calibration note:** chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9

**Experiment-id:** 20260118161538

**Experiment date:** 2026-01-18 16:15:38

**Experiment note:** temporary note!!!

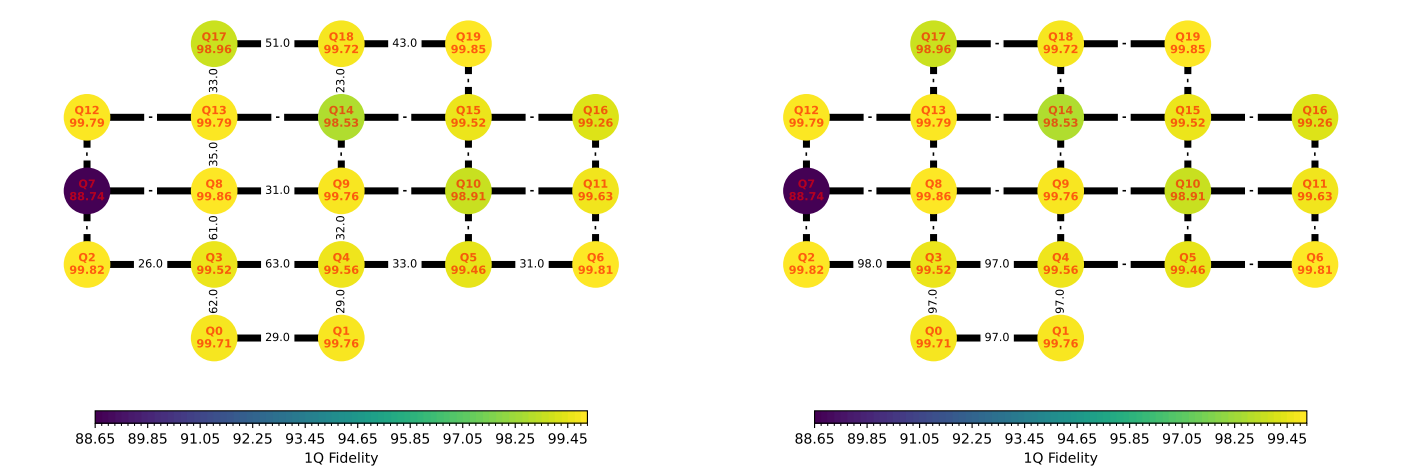
## 2. Version Comparison

| Library      | Version | Library | Version |
|--------------|---------|---------|---------|
| qibo         | 0.2.23  | numpy   | 2.4.3   |
| qibolab      | 0.2.9   | qibocal | 0.2.5   |
| matplotlib   | 3.10.8  | scipy   | 1.17.1  |
| scikit-learn | 1.8.0   | pandas  | 2.3.3   |
| networkx     | 3.6.1   | sympy   | 1.14.0  |
| torch        | 2.10.0  |         |         |

| Library      | Version | Library | Version |
|--------------|---------|---------|---------|
| qibo         | 0.2.22  | numpy   | 2.2.6   |
| qibolab      | 0.2.9   | qibocal | 0.2.4   |
| matplotlib   | 3.10.3  | scipy   | 1.15.3  |
| scikit-learn | 1.6.1   | pandas  | 2.2.3   |
| networkx     | 3.4.2   | sympy   | 1.14.0  |
| torch        | 2.7.0   |         |         |

## 3. One and two qubit fidelities

The single qubit fidelity is obtained via Randomized-Benchmarking. The two-qubit fidelity is the "Bell-state fidelity".



#### 4. Statistics

|                | Average | Median | Min   | Max   |
|----------------|---------|--------|-------|-------|
| T1 (ns)        | -       | -      | -     | -     |
| T2 (ns)        | -       | -      | -     | -     |
| Fidelity       | 0.928   | 0.999  | 0.642 | 1     |
| RO Fidelity    | 0.5     | 0.5    | 0.497 | 0.502 |
| 2q RB Fidelity | -       | -      | -     | -     |
| 2q CZ Fidelity | -       | -      | -     | -     |

|                | Average  | Median   | Min   | Max      |
|----------------|----------|----------|-------|----------|
| T1 (ns)        | 1.28e+04 | 1.23e+04 | 646   | 3.65e+04 |
| T2 (ns)        | 2.36e+25 | 3.68e+03 | 125   | 9.43e+26 |
| Fidelity       | 0.99     | 0.997    | 0.887 | 0.999    |
| RO Fidelity    | 0.794    | 0.777    | 0.777 | 0.927    |
| 2q RB Fidelity | 0.965    | 0.972    | 0.923 | 0.981    |
| 2q CZ Fidelity | 0.975    | 0.974    | 0.967 | 0.991    |

#### 5. Best Qubits Selection

| k-qubits | Best Qubits    | Fidelity |
|----------|----------------|----------|
| 2        | 3, 4           | 0.628    |
| 3        | 0, 3, 4        | 0.626    |
| 4        | 0, 3, 4, 8     | 0.621    |
| 5        | 0, 3, 4, 8, 13 | 0.554    |

| k-qubits | Best Qubits   | Fidelity |
|----------|---------------|----------|
| 2        | 2, 3          | 0.981    |
| 3        | 0, 2, 3       | 0.976    |
| 4        | 0, 1, 2, 3    | 0.970    |
| 5        | 0, 1, 2, 3, 4 | 0.965    |

#### 6. Summary metrics

| Benchmark       | Metric   | Value | Runtime    |
|-----------------|----------|-------|------------|
| Mermin          | Max      | 1.798 | 153.78 sec |
| Grover 2q       | Max prob | 0.938 | 2.97 sec   |
| Grover 3q       | Max prob | 0.818 | 2.08 sec   |
| GHZ state prep. | Success  | 0.945 | 2.84 sec   |

| Benchmark       | Metric   | Value  | Runtime  |
|-----------------|----------|--------|----------|
| Mermin          | Max      | -3.316 | 1.02 sec |
| Grover 2q       | Max prob | 0.89   | 2.36 sec |
| Grover 3q       | Max prob | 0.385  | 2.43 sec |
| GHZ state prep. | Success  | 0.910  | 3.95 sec |

#### 7. Randomized Benchmarking Results

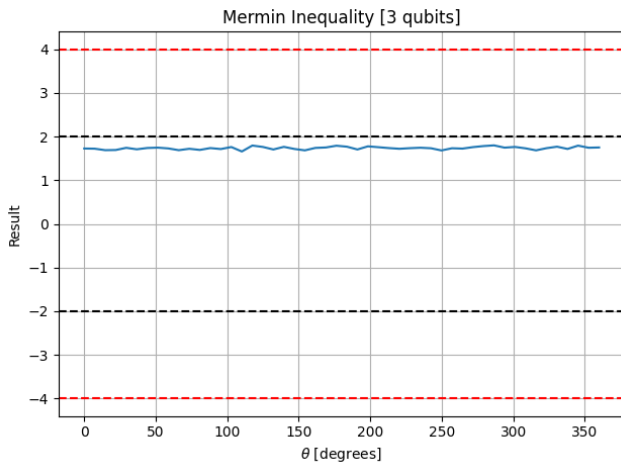
| Qubit n | Fidelity | Error Bars     | Qubit n | Fidelity | Error Bars     |
|---------|----------|----------------|---------|----------|----------------|
| 0       | 0.997    | $\pm 0.000131$ | 10      | 0.989    | $\pm 0.000582$ |
| 1       | 0.998    | $\pm 7.47e-05$ | 11      | 0.996    | $\pm 0.000218$ |
| 2       | 0.998    | $\pm 7.57e-05$ | 12      | 0.998    | $\pm 0.000126$ |
| 3       | 0.995    | $\pm 0.000163$ | 13      | 0.998    | $\pm 7.15e-05$ |
| 4       | 0.996    | $\pm 0.000185$ | 14      | 0.985    | $\pm 0.00183$  |
| 5       | 0.995    | $\pm 0.000147$ | 15      | 0.995    | $\pm 0.000469$ |
| 6       | 0.998    | $\pm 8.34e-05$ | 16      | 0.993    | $\pm 0.000558$ |
| 7       | 0.887    | $\pm 0.0287$   | 17      | 0.99     | $\pm 0.000502$ |
| 8       | 0.999    | $\pm 4.48e-05$ | 18      | 0.997    | $\pm 0.00026$  |
| 9       | 0.998    | $\pm 7.2e-05$  | 19      | 0.998    | $\pm 9.05e-05$ |

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|---------|----------|----------------|---------|----------|----------------|
| 0       | 0.997    | $\pm 0.000131$ | 10      | 0.989    | $\pm 0.000582$ |
| 1       | 0.998    | $\pm 7.47e-05$ | 11      | 0.996    | $\pm 0.000218$ |
| 2       | 0.998    | $\pm 7.57e-05$ | 12      | 0.998    | $\pm 0.000126$ |
| 3       | 0.995    | $\pm 0.000163$ | 13      | 0.998    | $\pm 7.15e-05$ |
| 4       | 0.996    | $\pm 0.000185$ | 14      | 0.985    | $\pm 0.00183$  |
| 5       | 0.995    | $\pm 0.000147$ | 15      | 0.995    | $\pm 0.000469$ |
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| 8       | 0.999    | $\pm 4.48e-05$ | 18      | 0.997    | $\pm 0.00026$  |
| 9       | 0.998    | $\pm 7.2e-05$  | 19      | 0.998    | $\pm 9.05e-05$ |

## 8. Mermin

Mermin's algorithm for 3 qubits.

| Runtime    | Qubits  | Max   |
|------------|---------|-------|
| 153.78 sec | 0, 3, 4 | 1.798 |



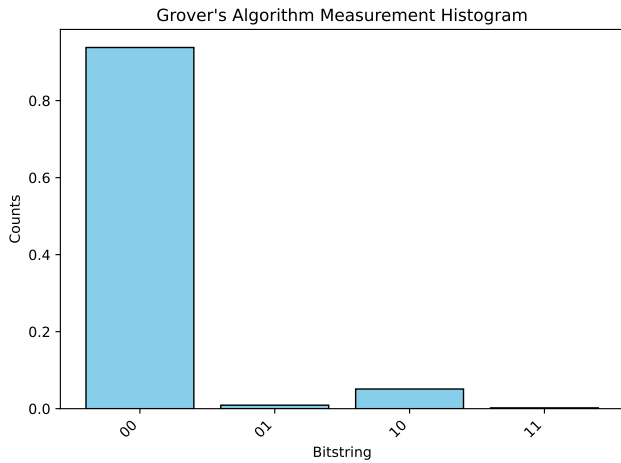
| Runtime  | Qubits  | Max    |
|----------|---------|--------|
| 1.02 sec | 0, 2, 3 | -3.316 |



## 9. Grover - 2 qubits

Grover's algorithm for 2 qubits executed on `sq20` backend with 1000 shots per circuit. We measure the success rate of finding the target state '11' for each pair of qubits in `[[3, 4]]`.

| Runtime  | Qubits | Max prob |
|----------|--------|----------|
| 2.97 sec | 3, 4   | 0.938    |



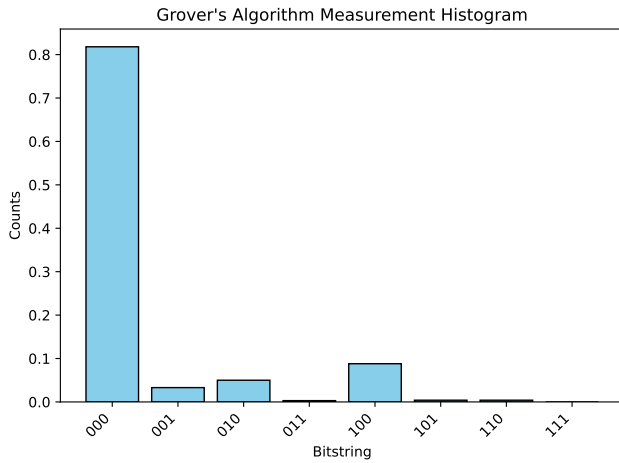
| Runtime  | Qubits | Max prob |
|----------|--------|----------|
| 2.36 sec | 2, 3   | 0.89     |



## 10. Grover - 3 qubits

Grover's algorithm for 3 qubits executed on sinq20 backend with 1000 shots per circuit. We measure the success rate of finding the target state '111' for each pair of qubits in  $[[0, 3], [3, 4]]$ .

| Runtime  | Qubits  | Max prob |
|----------|---------|----------|
| 2.08 sec | 0, 4, 3 | 0.818    |



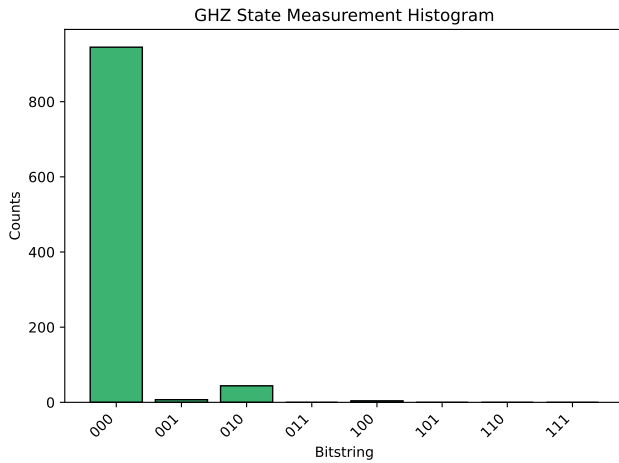
| Runtime  | Qubits  | Max prob |
|----------|---------|----------|
| 2.43 sec | 0, 2, 3 | 0.385    |



## 11. GHZ state preparation

GHZ circuit with 3 qubits executed on sinq20 backend with 1000 shots.

| Runtime  | Qubits  | Success |
|----------|---------|---------|
| 2.84 sec | 0, 3, 4 | 0.945   |



| Runtime  | Qubits  | Success |
|----------|---------|---------|
| 3.95 sec | 0, 3, 2 | 0.910   |

